

Practical #7

Practical Title:

To understand automated testing concepts and Selenium architecture for web applications

Theory:

1. What is Automated Testing?

Automated testing is a technique where test cases are executed using tools and scripts instead of manual effort.

It helps in:

- Faster testing
 - Repeated execution
 - Improved accuracy
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2. What is Selenium?

Selenium is an open-source tool used for automating web applications for testing purposes.

It supports multiple languages like:

- Python
 - Java
 - C#
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3. Selenium Components

Selenium IDE:

- Browser extension
- Record and playback testing

- Easy for beginners
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Selenium WebDriver:

- Core component
 - Directly interacts with browser
 - Supports automation using code
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Selenium Grid:

- Used for parallel testing
 - Runs tests on multiple machines/browsers
 - Saves time
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4. Selenium Architecture

Selenium architecture consists of:

- **Test Script** → Written in language (Python/Java)
- **WebDriver API** → Sends commands
- **Browser Driver** (ChromeDriver, GeckoDriver)
- **Browser** → Executes actions

Flow:

Test Script → WebDriver → Browser Driver → Browser

Procedure:

Step 1: Install Selenium and required dependencies

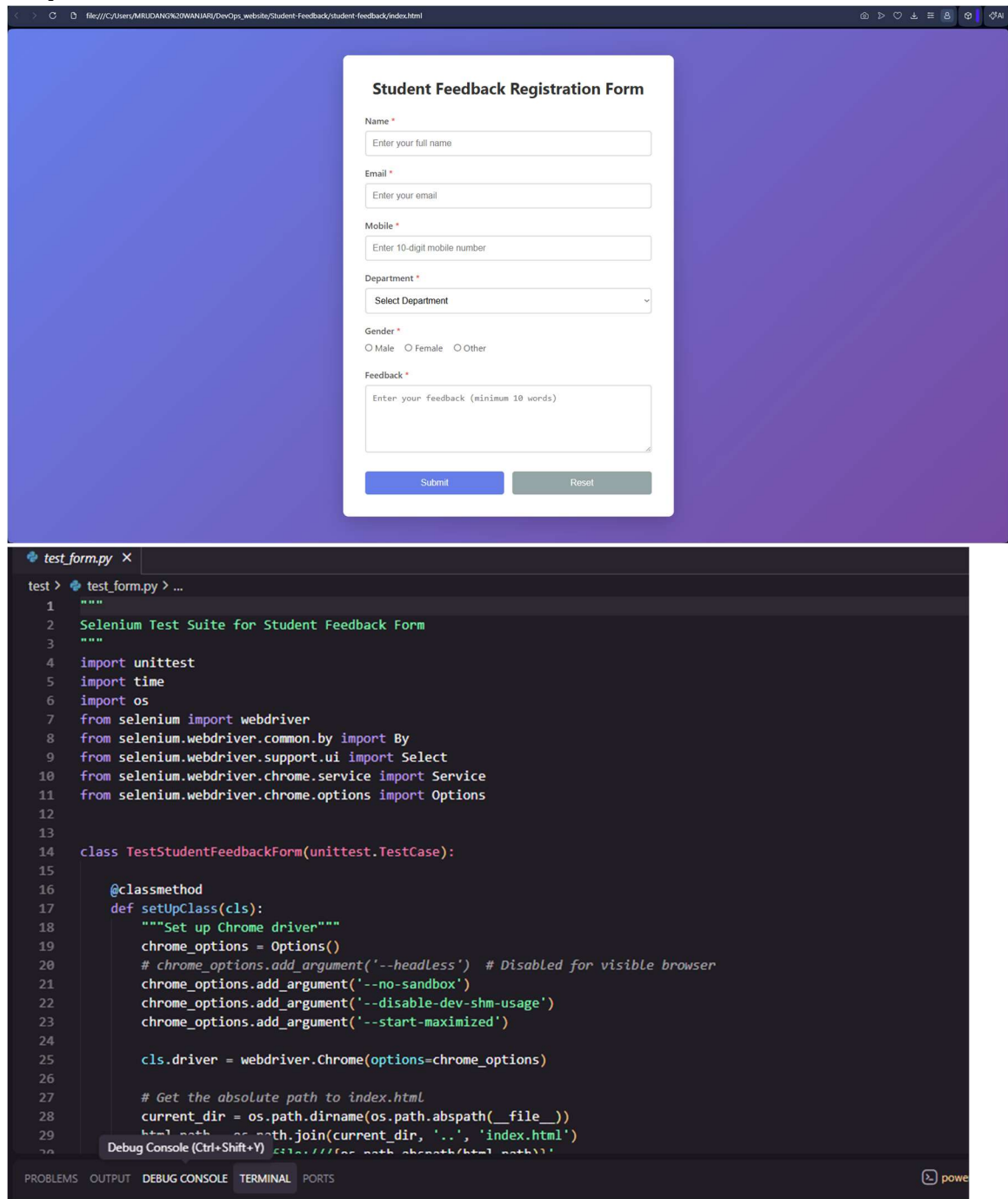
Step 2: Setup WebDriver

Step 3: Write a simple test script

Step 4: Perform actions (click, input, etc.)

Step 5: Execute the test script

Output Screenshots:



Activity / Task:

Task 1: Selenium Script

Code (Python Example):

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
import time

# Launch browser
driver = webdriver.Chrome()

# Open website
driver.get("https://www.google.com")

# Find search box and enter text
search_box = driver.find_element(By.NAME, "q")
search_box.send_keys("DevOps")
search_box.send_keys(Keys.RETURN)

# Wait for results
time.sleep(3)

# Close browser
driver.quit()
```

Task 3: Advantages of Automated Testing over Manual Testing

- Faster execution
 - Reusable test scripts
 - More accurate (less human error)
 - Supports continuous testing (CI/CD)
 - Saves time in long run
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Advantages

1. Fast and efficient testing
2. Supports multiple browsers
3. Reduces manual effort
4. Reusable scripts

5. Integration with CI/CD

Challenges

1. Initial setup complexity
2. Requires programming knowledge
3. Maintenance of test scripts